Skyshot Solver: Blind astrometry with nested stellar dominance levels and compact local descriptors

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Skyshot · https://skyshot.app

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Abstract

Blind astrometric calibration must identify an unknown star field despite unknown position, angular scale, orientation and image parity. Skyshot Solver addresses this problem by coupling a nested hierarchy of stellar *dominance levels* to compact local descriptors and direct hypothesis generation. A star belongs to a level when no brighter catalogue star lies within that level's angular radius. The same selection rule is applied to image detections over a ladder of pixel radii. Each indexed configuration consists of an anchor, a reference star and three neighbouring stars; a learned 32-bit descriptor represents its normalized geometry, while an 8-byte record retains the anchor identity and absolute reference-pair geometry. A key match therefore supplies a candidate centre, scale, rotation and parity without first recovering all constituent correspondences. Agreement between distinct anchors is tested before catalogue verification and distortion refinement. The proposed contribution is the integration of reproducible density selection, compact retrieval and pose-bearing records, rather than scale invariance or geometric hashing in isolation. In an author-run comparison on 91 development images spanning 0.381–20°, the evaluated implementation identified all fields and met a 3-pixel reference-grid criterion on every image, with median process-to-result latency of 29.7 ms. Six other solvers identified 7–89 fields under the tested configurations. We explain the mechanisms behind the observed advantages, isolate the incremental contribution of the learned descriptor, and state the evaluation's limits, including dataset reuse, multiple measurement sessions and processor wake-state differences.

Keywords: blind astrometry; star identification; geometric hashing; local descriptors; density normalization; statistical verification.

1 Problem and contribution

A blind plate solver estimates a world coordinate system (WCS), mapping image pixels to celestial coordinates, without a pointing, angular-scale, rotation or parity hint. This is particularly demanding for amateur astrophotography: catalogue depth, limiting magnitude, field width, optical distortion and the proportion of spurious detections vary greatly between instruments and processed images. A useful solver must retrieve the correct field quickly and reject plausible but incorrect alignments.

Skyshot Solver is designed around a specific observation: *local brightness dominance can select comparable stellar neighbourhoods in the catalogue and in the image before the image scale is known*. The unknown scale is handled through nested selection levels and a scale-normalized key. Absolute geometry is then recovered from the matched index record. This separates three tasks that otherwise interact expensively: choosing a useful density of stars, retrieving local geometric candidates, and evaluating a complete field hypothesis.

The methodological contribution has three linked parts:

1. **A reproducible hierarchy of stellar samples.** Dominance levels are nested, adapt to local stellar density and remain valid for retained stars under a pure faint-end magnitude cut. Image-side pixel levels can therefore query a common catalogue index without knowing which angular level is appropriate in advance.

2. **Compact retrieval with absolute geometry attached.** A 32-bit local key retrieves an 8-byte record carrying the information needed for a complete similarity hypothesis. The key describes a five-star configuration associated with an anchor; it is not a descriptor computed from an isolated star.
3. **Consensus before expensive verification.** Candidate hypotheses vote in position, scale, rotation and parity. Only sufficiently consistent votes from distinct anchors lead to catalogue extraction, statistical acceptance and WCS refinement.

The paper describes the core method in the implementation evaluated on 11 October 2026 (revision `50334ec`). The learned-key ablation uses an earlier fixed revision, `73106a3`. Later additions for panoramas, extreme wide fields and deeper companion indexes are discussed separately; their performance is not attributed to the historical comparison. The software and experiment records are author-maintained research artefacts [16, 17, 19].

2 Relation to established methods

Astrometry.net already demonstrates fully blind calibration through scale-invariant geometric codes, indexed star patterns and Bayesian verification [1]. Consequently, neither solving without a scale hint nor using geometric hashing is, by itself, a novelty claim for Skyshot. ASTAP constructs local four-star patterns and compares normalized inter-star distances within its catalogue-search procedure [2]. Watney uses a related quad-database approach [3]. TETRA and its descendants use compact four-star geometric hashes and were developed principally for star-tracker applications [4, 5, 6, 7]. These methods establish strong precedents for efficient local-pattern identification.

seiza is an especially relevant comparison because its documented blind solver also uses an all-sky pattern index, hypothesis voting and verification [8]. Voting therefore cannot be presented as unique to Skyshot. The distinction is the construction of the candidate population and records: a nested dominance hierarchy, depth-conditioned neighbourhood variants and an anchor-associated descriptor whose record preserves absolute basis geometry.

The dominance rule also has a mathematical precedent. Applied to a Poisson process with independent continuous marks, selecting points whose mark is locally minimal is Matérn type-II thinning [9]. Skyshot's contribution is its use as a catalogue–image selection interface in a blind astrometric system, together with the retrieval and validation design described below. Similarly, summation over shared per-neighbour features is a standard way to obtain permutation-invariant set representations [10]. Learned identification from local stellar graphs has also been demonstrated by SIGNN [11]. Skyshot uses learning to produce a compact retrieval key whose hits remain subject to geometric verification, rather than assigning one output class to every catalogue star.

3 Dominance levels as a common selection rule

3.1 Definition and deterministic properties

Let catalogue star i have unit direction $\boldsymbol{s}_i \in \mathbb{S}^2$ and brightness rank b_i, with smaller rank meaning brighter. Ties are broken deterministically by catalogue order. Its dominance radius is

$$D_i = \min_{j:b_j<b_i} \arccos(\boldsymbol{s}_i \cdot \boldsymbol{s}_j), \tag{1}$$

with $D_i = +\infty$ for the brightest star. For angular radii $R_{m+1} = R_m/2$, define

$$L_m = \{i : D_i \geq R_m\}, \qquad R_m = 1.94' \, 2^{7-m}. \tag{2}$$

The evaluated full index uses levels $m = 1, \ldots, 8$; the radius formula also defines coarser construction levels. Members of a given level are separated by at least R_m: of any closer pair, the fainter member would

Table 1: Architectural distinctions relevant to blind solving. Descriptions refer to the cited methods and the tested software families; they are not claims that other solvers cannot be extended.

Method family	Candidate construction	Distinction emphasized in Skyshot
Astrometry.net	Geometric pattern codes with indexes across angular scales; probabilistic verification.	Reproduce nested local-density selections and query one common key space; recover absolute scale from record payload.
ASTAP / Watney	Local quads and normalized edge ratios, with catalogue-region and scale search.	Let descriptor hits generate pointing and scale hypotheses directly, before extracting a verification catalogue.
TETRA / tetra3 family	Four-star hashes with fast database lookup, designed principally for star trackers.	Combine deeper catalogue sampling, variable image depth and a distortion-refined WCS for heterogeneous astronomical images.
seiza	All-sky pattern retrieval, pose voting and verification.	Dominance-based sample hierarchy and depth variants, plus learned anchor-associated keys and compact absolute-geometry payload.

be excluded. A star can belong to several levels, and several neighbourhood records can be attached to the same anchor.

Proposition 1 (Nesting and faint-end truncation)**.** *The levels satisfy $L_m \subseteq L_{m+1}$. If a catalogue is restricted to all stars brighter than a fixed magnitude limit, the dominance radius of every retained star is unchanged.*

Proof. The first statement follows from $R_{m+1} < R_m$. For the second, the restriction removes only stars fainter than a retained star, while the minimum in Eq. 1 ranges only over brighter stars. Its candidate set is unchanged. □

This property gives a practical way to model varying exposure depth. Restricting a dominance level to a magnitude cap reproduces the level of a catalogue truncated at that cap. It does *not* imply that the nearest-neighbour configuration is unchanged: fainter neighbours may disappear. Skyshot therefore stores additional configurations at magnitude caps 18, 16.5, 15, 13 and 11, adding records when the selected neighbour set changes. This spends index space on depth changes rather than assuming that a deep and a shallow image contain the same neighbourhood.

3.2 Why dominance tends to regularize density

Under a homogeneous Poisson model with parent intensity λ per steradian and independent continuous brightness marks, the retained intensity at radius R is

$$\lambda_R = \lambda \int_0^1 \exp(-\lambda A_R t)\, dt = \frac{1 - \exp(-\lambda A_R)}{A_R}, \qquad A_R = 2\pi(1 - \cos R). \tag{3}$$

Here t is the percentile of the star's mark and A_R is the area of a spherical cap. Equation 3 is the spherical-cap counterpart of the usual planar Matérn-II intensity [9]. For small R and a sufficiently dense parent catalogue,

$$A_R \simeq \pi R^2, \qquad \lambda_R \simeq \frac{1}{\pi R^2}. \tag{4}$$

Thus halving the radius approximately quadruples the selected density. In sparse regions, by contrast, $\lambda_R \simeq \lambda$: the procedure cannot create missing stars. Real stellar positions and brightnesses are neither homogeneous nor independent, so these expressions explain the normalization mechanism rather than prove uniformity of the real sky.

3.3 Selection in the image

Image detections are ordered by measured brightness and assigned pixel dominance distances d_i by the same nearest-brighter rule. Candidate pools use a quarter-octave pixel-radius ladder,

$$r_k = 2^{k/4} \text{ pixels}, \qquad P_k = \{i : d_i \geq r_k\}. \tag{5}$$

Pools containing roughly 10–800 stars are considered, beginning near 150 stars; duplicate pools are skipped. Under a local similarity of scale s radians per pixel, unchanged star identities and unchanged brightness order, $D_i \simeq s d_i$. A compatible pair of image and catalogue levels can therefore expose corresponding neighbourhoods without an initial value of s.

The qualification matters. Saturation, passband differences, extinction, missing brighter detections and an image boundary can change the rank order or censor the nearest brighter star. Dominance selection is robust through redundant configurations, depth variants and tolerant matching; it is not invariant to these effects. In particular, faint-end truncation is an exact catalogue property, while its transfer to a photograph is conditional on what the detector actually sees.

3.4 Configuration generation and search order

The first search pass prioritizes *dominant anchors*, for which $d_A \geq 2r_k$. The disc reaching an anchor's fourth nearest neighbour must fit within the image boundary. For each anchor, the first pass uses either of the two nearest neighbours as reference and triples drawn from the other four nearby neighbours. A second pass expands to three possible references and triples from five neighbours; a final exhaustive pass includes non-dominant anchors. Configurations already evaluated at another pixel level are skipped.

Catalogue construction uses a corresponding economical record plan. For dominant anchors, with the nearest neighbour as reference, the three-star subsets of the next four neighbours give four records at levels $m \leq 6$; only two are retained at level 7 and one at level 8. Non-dominant anchors receive one record at levels $m \leq 6$. Depth-conditioned variants are added as described above. This enumeration is bounded and selective; the method does not store or query every five-star combination in a field.

4 A compact descriptor tied to an anchor

4.1 Normalized local geometry

An indexed configuration contains an anchor A, a reference neighbour R and three additional neighbours S_1, S_2, S_3. Catalogue positions are projected onto the tangent plane at A; image positions are expressed in pixels. Write $\ell = \|R - A\|$, $\boldsymbol{e}_x = (R - A)/\ell$ and let $\boldsymbol{e}_y$ be the perpendicular unit vector. The normalized coordinates are

$$u_j = \frac{(S_j - A) \cdot \boldsymbol{e}_x}{\ell}, \qquad v_j = \frac{(S_j - A) \cdot \boldsymbol{e}_y}{\ell}. \tag{6}$$

Translation, positive uniform scale and in-plane rotation cancel. Reflection is canonicalized using

$$q = \sum_{j=1}^{3} \frac{v_j}{u_j^2 + v_j^2}, \qquad v_j \leftarrow -v_j \text{ if } q < 0, \tag{7}$$

while the applied flip is retained as a parity bit. Near $q = 0$, centroid noise can change the canonical branch; probing can include both branches. Degenerate configurations with a zero basis length or coincident neighbour are rejected.

The construction is invariant under the stated planar similarities, for a fixed configuration and ideal arithmetic; reflection canonicalization additionally excludes the degenerate case $q = 0$. It is not exactly invariant

under arbitrary perspective changes, optical distortion, affine anisotropy or fisheye projection. Those effects are handled through local configurations and subsequent geometric refinement, not by asserting a stronger invariance of the descriptor.

4.2 Learned quantization and uncertainty-guided probes

The evaluated default key, LGQ-GD, uses geometric and dominance features: log radius, angular harmonics, periodic functions of log radius and clipped log dominance-to-basis ratios. Measured magnitudes do not enter the learned key directly; brightness still determines the dominance selection. A shared multilayer perceptron processes each of the three neighbours, and its outputs are summed:

$$z = \rho\left(\sum_{j=1}^{3} \phi(\boldsymbol{f}_j), \boldsymbol{g}\right) \in \mathbb{R}^8, \tag{8}$$

where $\boldsymbol{g}$ contains anchor and reference dominance features. The sum makes the representation independent of the ordering of the three neighbours, following the standard set-aggregation construction [10]. The implementation has 32,424 network parameters, including an image-noise head, and eight empirical quantile tables of 4,097 entries each [18].

Each embedding coordinate is transformed by its empirical cumulative distribution F_d and quantized. With bit allocations $(4, 3, 3, 3, 4, 5, 5, 5)$, the packed key is

$$\begin{aligned} c_d &= F_d(z_d), \quad a_d = \min\{2^{b_d} - 1, \lfloor 2^{b_d} c_d \rfloor\}, \\ K &= \mathrm{concat}_{d=1}^{8}\left[a_d \ \mathtt{xor}\ (a_d \gg 1)\right], \qquad \sum_d b_d = 32. \end{aligned} \tag{9}$$

Gray coding makes adjacent quantization bins differ in one bit. The image-side noise model estimates boundary-crossing costs; the default solver enumerates eight low-cost probes, using subsets of the 12 least certain bits and, when warranted, the alternate parity branch. A key is therefore a retrieval device with multiple plausible quantizations, not an acceptance decision.

The training study used simulated fields sampled over 0.3–60°, with spatially separated HEALPix training, validation and test regions (70/10/20%) [13]. Configurations were admitted only when all five stars belonged to the same split. The pair-generation study produced 3.61 million matching pairs, comprising approximately 3.21 million training and 0.403 million validation pairs, together with six million catalogue samples [18]. The objective combines a Laplace negative log-likelihood for paired CDF residuals, marginal distribution regularization, decorrelation and a joint-uniformity term:

$$\mathcal{L} = \mathcal{L}_{\text{pair}} + \alpha\mathcal{L}_{\text{marginal}} + \beta\mathcal{L}_{\text{decorrelation}} + \gamma\mathcal{L}_{\text{uniformity}}. \tag{10}$$

The design balances stability under image perturbations against efficient use of key space. The real reference set nevertheless influenced feature selection, and the final probe budget was selected using development data. This is a learned retrieval component evaluated with additional holdout checks, not an entirely untouched end-to-end test.

5 From an index hit to a field hypothesis

5.1 Record format and lookup

The key space is shared by the indexed angular levels. A bijective 32-bit mixing function splits a key into a 20-bit bucket and a 12-bit fingerprint. The bucket-offset table locates a contiguous run of records, which is searched for the fingerprint. The split preserves the full mixed key: the 12-bit fingerprint alone is not the entire matching condition. Unrelated geometric configurations can still share a quantized 32-bit key.

Table 2: The 64-bit record in the evaluated full index. The bucket is implicit in the record's location. This is a size per configuration record, not per catalogue star or per complete index.

Field	Bits	Purpose
Key fingerprint	12	Completes the key match within the selected bucket.
Anchor identifier	24	Retrieves the anchor's absolute sky position.
Reference-pair length	10	Encodes $\log_2(\alpha_{AR}/R_m)$ at 1/256 increments, for angular separation α_{AR}.
Reference-pair angle	10	Encodes the pair's position angle over a full turn.
Parity	1	Records the canonical reflection choice.
Level	3	Identifies the angular dominance radius.
Dominant flag	1	Marks the stronger anchor-selection class.
Anchor dominance code	3	Supplies a coarse consistency filter.
Total	64	8 bytes.

The full evaluated index occupies about 0.88 GB in decimal units, including records, offsets and catalogue tables; these data are memory-mapped. Its source catalogue is ASTAP D80, derived from Gaia DR3 [2, 12]. Memory mapping reduces copying and enables sparse access, but it does not eliminate storage latency, page faults or the resident working set. The published latency measurements use a warm operating-system file cache.

5.2 Direct hypothesis generation

A hit supplies the catalogue anchor position, quantized angular reference-pair separation α_{AR} and position angle, level and parity. For the gnomonic catalogue basis, $\alpha_{AR} = \arctan \ell_{\mathrm{cat}}$. Combined with the image pair, these determine the candidate local angular scale $\hat{s} = \widehat{\alpha}_{AR}/\ell_{\mathrm{img}}$, orientation and reflection. The image anchor's offset then determines the candidate frame centre. Schematically, in a local tangent-plane approximation,

$$\boldsymbol{y} = \boldsymbol{t} + \hat{s}\, Q_{\theta,p}\boldsymbol{x}, \qquad \boldsymbol{t} = \boldsymbol{y}_A - \hat{s}\, Q_{\theta,p}\boldsymbol{x}_A. \tag{11}$$

The hypothesis has four continuous similarity parameters (two translations, one scale and one rotation) and a discrete parity. The implementation transports the orientation on the celestial sphere. The result is an approximate, quantized proposal, not yet a valid WCS.

This is the role of retaining absolute geometry beside an invariant key: normalization makes retrieval possible before scale is known, while the payload restores the geometric quantities needed for verification. A hit need not first expand into explicit correspondences for all five stars. Other pattern solvers also recover transformations from matched patterns; the advantage sought here is the compactness and low access cost of this particular representation.

5.3 Consensus and early rejection

Cheap consistency filters reject implausible field widths, inconsistent image and catalogue levels, and incompatible dominance intervals. Surviving hypotheses vote in frame-centre, log-scale, orientation and parity bins. In the evaluated configuration, coarse bins use approximately a quarter-field centre scale, 0.25 in log-scale and 10° in orientation.

A bin is considered only after evidence from at least two distinct image anchors. Before accessing a verification catalogue, a finer gate checks agreement at approximately 3% of the field size, 0.04 in log-scale and 3° in rotation, with tolerances enlarged for wide fields and distinct catalogue anchors required. This ordering matters: a repeated quantization hit from one anchor is not equivalent to independent spatial support, and cheap consensus can discard a large population of accidental key matches.

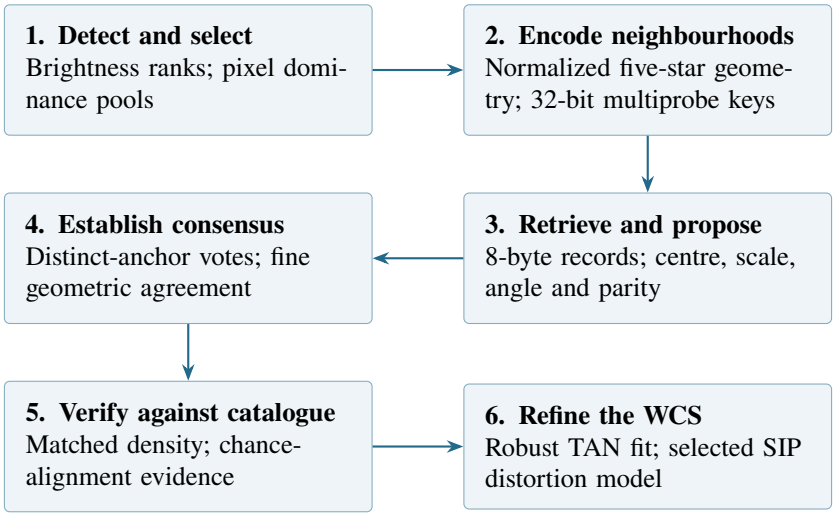


Figure 1: The core solving path. Catalogue retrieval for verification follows local-key retrieval and geometric consensus. Rejected candidates return to the search; accepted fields proceed to refinement.

6 Statistical verification and distortion refinement

The first verification stage fits a similarity from the supporting anchor pairs and compares image stars to a catalogue level chosen for the hypothesized scale. For image dimensions W, H, it selects the finest available level m_v whose approximate expected population satisfies $\hat{s}^2 WH/(\pi R_{m_v}^2) \le 250$, and uses detections with $d_i \ge 0.8 R_{m_v}/\hat{s}$. Thus verification density is matched to the image rather than taken from the entire deep catalogue.

Let $N_{\mathrm{img}}, N_{\mathrm{cat}}$ be the selected image and catalogue populations and k_1 the first-stage match count. With $r_1 = \max\{4, 0.004 \max(W,H)\}$ pixels, define $p_1 = \min\{0.999, N_{\mathrm{img}} \pi r_1^2/(WH)\}$ and $a = k_1/N_{\mathrm{cat}}$. For $a > p_1$, the implementation uses a binomial Chernoff-bound screening score,

$$B = -N_{\mathrm{cat}} \left[a \log \frac{a}{p_1} + (1-a) \log \frac{1-a}{1-p_1} \right], \tag{12}$$

with the usual limiting convention for $a = 1$. It requires $k_1 \ge 5$ and $B \le -14 - \log N_v$, where $N_v \ge 1$ counts attempted verifications [16]. This is a logarithmic upper bound on a binomial tail, not an exact tail evaluation.

The second stage performs three robust TAN fits with shrinking matching radii. For a final matching radius r, the adopted uniform spatial-density model gives an expected accidental count $\lambda_0 = N_{\mathrm{img}} N_{\mathrm{cat}} \pi r^2/(WH)$. If k matches are found, the evidence score is the Poisson Chernoff exponent

$$E(k, \lambda_0) = k \log \frac{k}{\lambda_0} - k + \lambda_0, \qquad k > \lambda_0. \tag{13}$$

For a Poisson variable X of mean λ_0, $\Pr(X \ge k) \le \exp[-E(k, \lambda_0)]$. The acceptance threshold is approximately $E \ge 30 + \log N_v$, together with at least eight matches and spatial spread of at least 20% along both image axes. These are operational acceptance rules. The score is not a calibrated posterior probability that a field is correct; shared stars, structured artefacts and model fitting can violate independence assumptions. Accounting for attempted verifications is useful, but does not by itself establish a universal false-acceptance guarantee.

After identification, the solver fits a WCS in the standard celestial-coordinate framework [14] and models distortion using the SIP convention [15]. Model complexity is selected on held-out matched stars within the frame. For wide fields, progressively larger radial zones allow the model to grow from a reliable central fit; additional high-order radial terms can be folded into SIP. The refined model must recover at least 90% as many unique matches as the identification model when both are evaluated against the same deeper catalogue

at 2.5 pixels. This limits losses of support during refinement. It compares match counts, not retention of the exact same star pairs, and does not prove accuracy at every pixel.

The distinction between identification and refinement is central to the practical advantage. Finding the correct region of sky is insufficient for overlays, coordinate readout and image registration when unmodelled distortion displaces the corners. Conversely, a flexible distortion model must not be allowed to manufacture support for an incorrect field.

7 Where the computational advantages arise

The architecture targets the number of costly candidates rather than relying on one universally cheaper geometric primitive. Let N_c be the number of image configurations evaluated, K_p the probe budget, $\bar{b}$ the mean number of retrieved records per probe and N_v the number of hypotheses reaching catalogue verification. A useful decomposition is

$$\begin{aligned} T \approx\ & T_{\text{detect}} + T_{\text{select}} + N_c T_{\text{encode}} \\ & + N_c K_p (T_{\text{lookup}} + \bar{b}\, T_{\text{hit}}) + N_v T_{\text{verify}} + T_{\text{refine}}. \end{aligned} \tag{14}$$

This is a cost model, not a worst-case complexity bound. It makes the intended savings explicit:

- **Dominance sampling limits unproductive configurations.** Comparable local densities make it more likely that image and catalogue neighbourhoods share stars; nested pools avoid rebuilding an unrelated selection for every scale hypothesis.
- **Depth variants preserve useful configurations across exposure depth.** The exact truncation property justifies how catalogue samples are related, while storing changed neighbour sets supplies retrieval redundancy.
- **A compact payload makes each hit useful immediately.** The solver can reject a scale or position proposal before loading a larger catalogue region.
- **A more selective learned key reduces $\bar{b}$.** Its per-configuration evaluation costs more than the geometric baseline, but accidental collisions and record reads are reduced.
- **Distinct-anchor consensus reduces N_v.** Multiple geometrically consistent proposals carry more information than many near-duplicate hits from one neighbourhood.

These mechanisms explain a plausible advantage for blind, heterogeneous imagery. They do not imply that every individual component is necessary, that every competitor performs more operations, or that the measured wall-time ratio can be attributed solely to the key. End-to-end timing also includes decoding, detection, process startup and storage access. The component ablation below provides narrower causal evidence for the learned key; equivalent ablations of every architectural component remain future work.

8 Experimental evidence

8.1 Data, reference and measurement protocol

The seven-solver comparison contains 91 development frames: 85 real astronomical images and six synthetic renders. They span 0.381–20° on the longer side, 0.58–45.5 arcsec/pixel and 0.7–16.8 megapixels. The real material includes public blind-solve frames, Digitized Sky Survey cut-outs, repeated exposures of M66 and NGC 6888, and user-submitted images. Five additional frames of 0.17–0.24° were excluded for *all* solvers because they lie below the evaluated Skyshot configuration's declared 0.25° lower limit. Their outcomes in earlier development experiments were already known, including three Skyshot non-solutions; the scope restriction was not outcome-blind or preregistered. The declared 0.25–60° operating interval is therefore broader than the range tested in this comparison.

Field identities were checked against a second identification; the 85 real reference WCSs were refitted to Gaia-derived D80 stars, while the six synthetic references retained simulator truth. The identifying sources

Table 3: Historical comparison on 91 development frames. Times are seconds over each solver's identified frames. Q is the median of paired per-frame ratios $t_{\text{other}}/t_{\text{Skyshot}}$ on their common identified subset, whose size equals "Id." here. The processor wake-state caveat in Section 8 applies.

Solver	Id.	≤ 3 px	Wrong	No sol.	Median	p95	Q
Skyshot	**91**	**91**	**0**	**0**	**0.0297**	**0.0517**	1.0
seiza	89	87	1	1	0.987	1.286	30.9
Watney	74	45	0	17	0.686	0.972	24.4
tetra3rs*	69	49	0	22	0.846	0.981	28.2
ASTAP	67	50	0	24	5.827	28.463	161.3
astrometry.net	66	48	0	25	20.548	39.158	814.4
cedar-solve*	7	7	54	30	0.491	0.504	15.7

*Star-tracker configurations require separate interpretation. cedar-solve used its shipped 10–30° Hipparcos database; most test images are outside that range. tetra3rs used a 1–30° database limited to bright stars ($G < 10$). Both adapters incurred process startup and Python centroid-extraction costs. These rows do not measure the maximum attainable speed or coverage of their underlying matching algorithms.

include ASTAP, an earlier Skyshot implementation, DSS headers and synthetic truth. This provides checked field references, but is not a catalogue-independent accuracy experiment: the final reference and Skyshot use related catalogue material. FITS, JPEG and PNG metadata were removed, input names were hashed and decoded pixels were checked. Watney received 16-bit conversions for 21 floating-point FITS images because of its input-format constraint.

Every timed solve started a new process on one Apple M5 Max with 18 cores, 128 GB RAM and macOS 27.0.1. Wall time includes process startup, index opening, image decoding, detection, solving and output. An untimed warm-up preceded measurement; file caches were warm. The nominal limit was 60 s, with additional termination slack for astrometry.net. Frames returning a solution on the first pass, including wrong-field outputs, were timed in three passes and summarized by their median; non-solutions were not timed three times.

The measurements took place in separate sessions on 11 October 2026. ASTAP and astrometry.net used the earlier idle-gate session; Skyshot was subsequently remeasured with an active waiting gate to avoid processor wake-up latency. The other four solvers were measured in a later session using the active gate. Skyshot's earlier median and p95 were 50.3 and 74.0 ms, versus 29.7 and 51.7 ms in the remeasurement used in Table 3; both runs satisfied the correctness criterion on all 91 frames. Thus the comparison shares the machine, input sanitation and scoring protocol, but does not provide a fully uniform processor wake-state experiment. Its large ratios should be interpreted as observed process-to-result ratios for these runs, not isolated algorithmic speedups [17].

An output is *identified* if its field centre is within 50 pixels of the reference. It meets the stricter *3-pixel criterion* if the maximum discrepancy on a 3×3 grid of the centre, edge midpoints and corners is at most 3 pixels. A returned solution failing identification is counted as a wrong-field output. The grid measures agreement with the reference at nine positions; it does not establish a uniform 3-pixel bound over the continuous image. Both pixel-row conventions were tested and the better reference agreement retained to accommodate differences between file readers. The star-tracker adapters returned matched pairs, from which a TAN WCS was fitted for this scoring.

8.2 Comparative outcomes

Skyshot identified 91/91 frames and met the stricter grid criterion on all of them. Its median grid discrepancy was 0.128 pixels and its 95th percentile was 0.712 pixels. seiza was closest in coverage, identifying 89/91 and meeting the 3-pixel criterion on 87/91. The observed difference is therefore not only latency: within this dataset and configuration, Skyshot combines broad identification coverage with distortion-refined reference agreement.

The field-width breakdown is consistent with this interpretation, though small at the extremes. Skyshot identified all six frames below 0.5° and all seven at or above 8°; seiza identified four and seven, respectively. Watney identified two and zero. These counts indicate where the tested configurations differed, without establishing a universal lower or upper field limit for the competing methods.

The paired ratios in Table 3 use

$$Q_j = \text{median}_{i \in C_j} \frac{t_{ij}}{t_{i,\text{Skyshot}}}, \tag{15}$$

where C_j contains frames identified by both solvers. They are not ratios of two independently computed medians, and the subsets differ between solvers. Ranking only successful-case latency would omit coverage; reporting identification and accuracy alongside time avoids that omission.

The cedar-solve results illustrate why database compatibility matters. All 54 returned outputs failing the identification test came from fields narrower than the intended database range. Within the seven approximately 10–20° frames, it identified all seven and satisfied the 3-pixel criterion. Its median internal extraction-plus-solve time there was about 29.8 ms, far below its approximately 491 ms process-to-result time. This prevents the process-level comparison from being read as evidence that its hash matcher is intrinsically slow.

8.3 Ablation: what the learned key adds

The end-to-end comparison cannot isolate the contribution of the descriptor. A separate fixed-binary A/B study changed the index key while retaining the record contract and solver [19]. It compared the handcrafted baseline key with LGQ-GD using eight probes. The reference experiment contains 96 images; its timing uses minima over repeated, alternating runs on a loaded machine. These values are therefore kept separate from the idle-machine process-level comparison.

Table 4: Key-only A/B evidence. Times are milliseconds and are taken from the documented component study, not the seven-solver benchmark. Holdout real images lack an independent reference WCS; their solutions are assessed by agreement between arms.

Measure	Handcrafted key	Learned key, $K_p = 8$
Reference images meeting the correctness test	93/96	93/96
Reference solver-stage median, all 96 frames	5.02	4.22
Reference full decode–detect–solve median, 93 common solves	15.64	14.36
Reference solver-stage p95, 93 common solves	27.1	18.0
Holdout real images returning a solution	113/114	113/114
Holdout real solver-stage median	5.00	4.20
Holdout real end-to-end median	16.6	15.4
Holdout real end-to-end p95	48.6	27.1
Accepted random negative lists	0/30,000	0/30,000
Accepted negative images	0/200	0/200

The learned key's advantage comes primarily from selectivity. The offline study reports 1.04–1.11 accidental hits per probe versus 6.0–13.6 for the geometric key, while retaining approximately 0.97 of its per-anchor true-hit recall on the real reference material [18]. In the final paired analysis, the median solver-stage time ratio was 0.836 on reference images and 0.840 on the real holdout; the corresponding end-to-end ratios were 0.939 and 0.913 [19]. This supports a roughly 16% reduction in solver-stage latency and a smaller 6–9% reduction in end-to-end latency under those conditions. It does not support attributing the full multi-solver speed difference to learning.

The holdout also contains 400 synthetic star lists: both keys solved 398 correctly with no wrong field. The 114 real holdout images are disjoint by file hash from the reference set, but 98 come from the same target session represented in the reference data. They provide additional evidence of parity and timing, with limited diversity and without independent absolute truth for their WCS.

No false accept was observed among 30,000 random negative lists or 200 negative images in either arm. For n independent trials from one fixed distribution and zero events, the one-sided 95% binomial upper bound is $1 - 0.05^{1/n}$: approximately 1.0×10^{-4} for the lists and 1.49% for the images. These conditional bounds do not transfer automatically to structured adversarial inputs or different exposure distributions. Repeated runs of the same negatives are not additional independent samples. The list experiment also used a 600 ms deadline, truncating the handcrafted key's search on approximately 37% of inputs.

9 Interpretation, limits and development beyond the comparison

Supported advantage. The evidence supports a concrete claim: in the reported blind-image experiment, the evaluated Skyshot configuration combined the highest coverage, complete satisfaction of the chosen reference-grid criterion and the lowest measured process-to-result latency. The method offers a coherent explanation through density-compatible selection, inexpensive pose proposals and delayed catalogue verification. The key ablation independently supports the narrower claim that improved retrieval selectivity reduces solver work without a measured loss of reference coverage.

Scope of originality. The distinctive proposal is the combination of the dominance hierarchy, depth-conditioned configurations, compact learned keys, absolute-geometry payload and consensus-first validation. Geometric hashing, invariant coordinates, local thinning, set aggregation and hypothesis voting each have prior art. This paper does not establish that the combination is the first of its kind across all published and unpublished systems, nor does the comparison establish superiority for every observing task.

Dataset and measurement limits. The seven-solver set was used during Skyshot development and includes repeated exposures of a few fields. It is not an external held-out benchmark. References partly originate from tested or related solvers, and use Gaia-derived catalogue data. Measurements use one machine, warm file caches and multiple sessions; the Skyshot remeasurement additionally changes the wake-up protocol. Confidence intervals from treating all frames as independent would understate shared-field dependence, so none are used to imply a population-wide ranking.

Configuration limits. Competing programs were run blind with fixed settings and installed databases. Position and scale hints, persistent workers, matched catalogue depth, alternative star detection, retuned distortion models or cold-cache operation may change the results substantially. astrometry.net did not have its smallest-scale 4200–4203 indexes installed. The numerical table describes these configurations, not the best achievable configuration of each solver. Detailed versions and settings are retained in Appendix A.

Method limits. Descriptor repeatability falls when brightness rankings or neighbourhood membership change. Wide-field projection and optical distortion violate the local similarity approximation; the learned-key study found weaker per-anchor recall at 40–60°, mitigated by additional search in the tested solver settings. Catalogue epoch and proper-motion errors can also matter; the evaluated D80 path has no per-star proper-motion propagation. The default evaluated search accepts the first verified field and does not prove global uniqueness over all possible solutions.

Later extensions. The repository subsequently adds landscape masking, local-window searches, companion indexes and additional projection models. These extend the same local-identification idea to scenes that cannot be represented well by one whole-frame similarity. They require further checks that an extended solution is supported beyond its seed window. Those developments are outside the seven-solver experiment and should be evaluated with their own versioned data and timing protocol [16].

The next decisive evaluation is a preregistered, externally assembled set stratified by distinct sky field, angular width, depth, distortion and non-stellar content. It should separate cold and warm caches, use matched processor wake conditions, test both fresh processes and persistent workers, and repeat with useful priors. Further component ablations should remove depth variants and consensus gates individually to quantify their contribution and interaction. These tests would strengthen the causal and generalization claims beyond the evidence available here.

10 Conclusion

Skyshot Solver organizes blind astrometry around a shared, nested notion of local stellar dominance. Scale-normalized neighbourhoods make retrieval possible before the field scale is known, while compact records restore the absolute geometry needed to propose a complete alignment. Agreement between distinct anchors concentrates expensive catalogue verification on coherent proposals, and distortion refinement converts identification into a usable WCS.

The resulting architecture achieves a strong combination of coverage, reference agreement and latency in the reported development-set comparison: 91/91 identified fields, 91/91 passing the 3-pixel grid test, and 29.7 ms median process-to-result time under the measured conditions. A separate ablation shows that the learned key contributes through fewer accidental retrievals, with approximately 16% lower median solver-stage latency and preserved reference coverage. These results motivate the method as an efficient approach to heterogeneous blind astrometry, with the specific benefits and boundaries made explicit rather than inferred from speed alone.

Competing interests. The author develops Skyshot Solver, one of the systems evaluated here. The experiments and interpretation are the author's own research evaluation.

A Tested configurations and reproducibility

Table 5: Versions, installed data and main blind-search settings of the historical comparison. Sizes are approximate decimal GB of the catalogue/index files used, not peak RAM.

Solver	Version	Data; size	Main configuration
Skyshot	`50334ec`	Gaia DR3 / D80-derived full index; 0.88	Default blind `solve`, LGQ-GD key.
ASTAP	CLI 2026.09.01	D80; 1.33	`-r 180 -fov 0 -z 0 -sip -wcs`.
astrometry.net	0.97	2MASS 4204–4219; 2.63	All installed indexes in parallel; downsample 2 above 2500 px; CPU limit 60 s.
seiza	0.25.1	Gaia DR3 blind $G \leq 16$ and verification $G \leq 17$; 3.17	Blind mode; scale 0.1–60 arcsec/pixel; SIP order 3.
Watney	2.0.1; `517aa3b`	Quad database v3, all parts; 4.81	Blind mode; radius range 0.101–16°; native macOS ARM64 build.
tetra3rs	0.14.0	Gaia DR3 $G < 10$, 1–30°; 1.75	No field-size filter; cedar-solve centroids; TAN fit to returned pairs.
cedar-solve	0.5.1	Shipped Hipparcos, 10–30°; 0.008	No FOV estimate; own centroider; TAN fit to returned pairs.

The numerical summaries can be recomputed from the author's archived per-run records and scripts. A complete experimental rerun additionally requires the original images, reference WCSs, catalogues and versioned solver binaries. The joined summary is `outputs/h2h-more/summary.json`; the original runs and environment descriptions are `outputs/h2h/runs.jsonl`, `outputs/h2h/environment.json`, `outputs/h2h-more/runs.jsonl` and `outputs/h2h-more/environment.json`. The harnesses are `bench/headtohead.py`, `bench/headtohead_more.py` and `bench/st_solve.py`. The component experiment is documented in `docs/learned/AB.md`, with inference details in `docs/learned/FEATURES.md`. The date-specific wake-state correction is recorded in `VALIDATION.md`.

The accompanying source package includes an evidence manifest with file digests and the numerical table used in this manuscript. These local records have not been assigned a public dataset DOI. Reference images include user-submitted material, so the paper makes no claim that every input image is publicly redistributable. A public replication package should use authorized images and immutable software/data identifiers.

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